Is what we see a projection? Objects as the common part of two points of view, tested on a quantum-Darwinism model

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Abstract

When two observers correctly read the same quantity and compare their records, they find them in agreement, within the measurement errors. This deposit proposes to read the object both of them see as the common part of their records : what each reads from its own record, without looking at the other's, with the same value as the other (the Gács–Körner common part, exact or with a very small tolerance when the copies are imperfect). The key word is projection: the reading projects a record onto a simpler question, like a shadow, and the projections of two records coincide, in the model within a chosen tolerance. In the author's framework the observed object is a transposition and non-local projection of the observer's "observation matrix"; in the model of this work the matrix is represented, by a choice of the AI, by the raw record of what the observer collected. What two projections share is the trace of one non-local relation, non-local in the sense of not located in space and time, not of an influence at a distance.

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Important before starting: what has been corrected in this version 2.62.1

Before getting to the heart of things, let us look at an important correction. The correction changes the way we read some results compared with the previous version (2.62).

The author asked for an independent check of the work. This check found some precise points in version 2.62. At those points the version presented some things as results about the author's thesis. Those things instead depended on translation choices made by the artificial intelligence (AI). The AI had not declared these choices. This version 2.62.1 corrects those points. It also adds the final section on an object that can have more values. The programs, the calculations and the numbers of version 2.62 do not change.

Here are the ten corrected points, explained one by one:

1. **The criterion on compared records (C1).** A **record** is everything that an instrument or a person has noted down in its own physical memory. Think of a laboratory notebook or of the memory card of a camera. The author's criterion C1 concerns only the records that are compared with each other. Version 2.62 had added a second part. This part was not in the author's criterion. The added part said a precise thing. If two observers have a very likely agreement even without comparing, then there must necessarily be a common object. Now this addition by the AI has been removed. Without it, two outcomes change. The reading "the same matter" (called T-1) is no longer excluded by the first criterion. It remains, however, "non-conforming" to criteria C4 and C5. Instead, the reading "the exact common part" (called T-4s) becomes compatible.

2. **The author's "matrix".** The word **matrix** appears in the author's text. The AI had rendered it in the model by the raw record of all the **photons** collected. A photon is a single grain of light. It is like a tiny drop in the rain of light that bounces off a chair and reaches our eyes. Some numbers say that the common part of the whole record is very small (8.5 % and 3.4 %). These numbers concern that choice of the AI, not the author's matrix.
3. **The common divisor of the two GCDs (T-3).** The **Greatest Common Divisor (GCD)** of two numbers is the largest number that divides both of them without remainder. For example, think of a pizza cut into 6 slices and one cut into 9 slices. Their greatest common divisor is 3. This formulation (T-3) is compatible as an arithmetic rule of the common part. It is in fact computed using only the **joint distribution** (test P1, already registered). A joint distribution is a single table. The table lists all the possible pairs of results for the two observers. It also says how likely each pair is. It is like a scoreboard that shows how many times two friends draw a red or a black card together. The criterion on distances (C5) is met only if it is used together with **triangulation**. The same happens for formulation T-4. Triangulation is the ordinary geometry used to find a distance by looking from two different points.
4. **The "nexus" and the copies (T-2 and T-5).** The author uses the word **nexus** for the link at the basis of what is observed. Identifying the "nexus" with the value of the source in the model had been a choice of the AI. So two readings cannot be judged within this model. The first is based on "copies of the same fact" (T-2). The second is based on "chains of copies" (T-5).
5. **The paradoxical side of the thesis.** What seems strange or paradoxical in the thesis concerns the **non-local link**. It is a link that is not in space and time. The paradoxical side does not concern the common part recorded in the memories.
6. **The "yes or no" behaviour and the steps.** The common part behaves like a "yes or no" switch, that is, all or nothing. This happens because in the first model the object is worth only **1 bit**. A bit is the smallest amount of information. It is what is needed to answer a question with only two equally likely choices. Think of "heads or tails" when you toss a coin. In chapter 7, instead, the object can take more values. There the common part grows in steps, within the class of readings tested.
7. **Distance from luminosity (T-7L) and fractionation.** Computing the distance from luminosity (T-7L) was not a "non-reciprocal" calculation, as 2.62 said. In reality this calculation needs only one thing. One needs to know the total number of copies emitted by the source. This number is a datum distinct from the mass. In the model it is not derived from the mass. Moreover, triangulation is ordinary geometry. It does not translate what the author calls "fractionation".
8. **The probabilities of quantum mechanics. Quantum mechanics** is the branch of physics that studies how very small things work. Think of atoms and of the grains of light (photons). In this work some definitions based on probabilities were tested. These definitions directly use the distributions of quantum mechanics. In the cases examined, they give no statistical prediction different from that of quantum mechanics.
9. **Mass and the static model.** In the **static** model, the definition of the common object is separate from mass. The model photographs everything in a single instant, frozen in time.
10. **The new chapter on the object with more values (chapter 7).** In the new chapter the object can be in more positions. In this case the common part grows in steps, within the class of readings tested. The copies can be locally blurred, that is, err towards the nearby cells. Then the coarse fractions appear first. However, the steps do not follow the exact divisors of the grid. Moreover, the common divisor of the two GCDs can have two roles. It can become the maximum ceiling or only an intermediate stage. It depends on where the **noise** enters. Noise is the disturbance that makes copies wrong, like the hiss in a phone call.

The summary in a few words

We are about to see the central idea of the whole work. We will also see what came out when it was put to the test with numbers.

Let us think of two **observers**, that is, two people or two measuring instruments. The two correctly read the same quantity. Then they compare their **records**, that is, the memories where they noted down the result. In this case they always find them in agreement, taking into account the normal small measurement errors. No case is known in which they find a systematic disagreement.

Here a precise way of reading the object that the two see is proposed. The object is **the common part of their records**. What does "common part" mean? It is what each reads in its own record, without peeking at the other's record. And each gets exactly the same value as the other. Suppose two deskmates look at the blackboard and take notes. Their notebooks will have different handwriting and different doodles in the margins. But the sentence written on the blackboard will be identical in both notebooks. That identical part is the common part.

This idea was put to the test. Before doing the calculations, eight different **mathematical formulations** were written and registered. They are eight ways of translating the idea into precise mathematical rules. Then the formulations were tested on a small model. The model is of the same type used by physicists in **quantum Darwinism**. Quantum Darwinism is a physical theory. According to this theory a thing becomes objective for everyone because the environment around it

makes very many copies of it, for example with light. It happens as with a piece of news, which becomes public when it is printed in millions of newspapers.

What was the outcome of the tests?

- The formulations that measure the common part exactly hold.
- Those that measure it with a small **tolerance** also hold. Tolerance means accepting a tiny probability of error. It is like saying "we agree almost always, except once in a million".
- Finally, those that measure it in a graded way hold.
- The formulation of the common divisor of the two GCDs also holds.
- The reading called "the same pattern" falls, instead.

In the model, the copies that reach the observers can be imperfect, that is, contain some errors. In that case the exact common part is zero until the two records are compared with each other. The exact common part instead becomes equal to the whole object at a precise moment. It is the moment when observer B faithfully copies the reading of observer A.

Still in the model, the object is worth 1 bit. This means that it has only two possible positions, like a coin that shows heads or tails. Here the common part with tolerance behaves like a "yes or no":

- it is zero if the observers read a number of copies below a certain threshold;
- it becomes about the whole object as soon as they go above that threshold.

The definitions based on probabilities use the same probabilities as quantum mechanics. In the cases examined, by construction, they give no different statistical prediction.

The key word of the whole text is **projection**. A projection is like the shadow of an object on the wall. The shadow keeps the outline but loses many inner details. Each observer sees the object through a projection of its own record. This projection is called a **reading**. The projections of the two records coincide (in the model, within the chosen tolerance). In the author's theoretical framework, what two projections have in common is the trace of one and the same **non-local** relation. It is a relation that is not located in space and time.

What happens with an object that has more values (chapter 7, added in this version). Imagine that the object does not have only two positions. It can be in 12 different positions, like the hours on a clock face. And it is read through blurred copies. Then, within the class of readings tested, the common part no longer arrives all at once. Instead, it grows in steps. First the observers agree on the halves, then on the thirds, then on the quarters and so on. They add one boundary at a time, within a few tens of copies. These steps, however, do not follow the exact divisors of the grid. Moreover, the observers use two rulers. The common divisor of the two rulers can become the maximum ceiling of the common part. Or it can be only a stage along the way. It depends on where the noise enters.

A guide to find your way through the chapters. Chapter 2 explains the thesis without using formulas. It is best to read it first. In particular, section 2.2 uses three concrete images to make the key word, projection, clear. Chapter 1 tells about the scientific studies that already exist. It helps to understand what other physicists say. If you want to go straight to the point, you can also skim it. Chapter 3 describes the model used for the tests. Chapter 4 shows all the results. It starts from three simple examples that can be checked by hand. Chapter 7 tells about the test with the object with 12 positions. It can also be read on its own.

1. What is already known in physics

Before explaining the proposal of this work, let us take a panoramic tour. Let us see what physicists have already discovered on a precise topic. The topic is how two observers agree when they look at the world.

Compared records always agree. Let us look at all the experiments of a certain type ever carried out. In these experiments two observers correctly read the same quantity. Then they compared their records. They always found them in agreement, within the normal margins of error of the instruments. This also holds in very special experiments. They are called experiments on "**Wigner's friend**".

What is the Wigner's-friend scenario? Imagine a scientist (the "friend") shut inside a laboratory. The laboratory is completely isolated from the rest of the world. It is like a safe from which not even a grain of light or a radio wave gets out. The friend makes a measurement on a particle. Then the friend writes the result in a notebook. Outside the laboratory there is another scientist, called Wigner or "superobserver". Wigner describes the whole closed laboratory as a single system. The system also contains the friend and the notebook. The system is governed by the laws of quantum mechanics. Suppose that the superobserver opens the door of the laboratory and asks the friend what was seen. In the

description of these experiments too, Wigner finds exactly what the friend recorded (study by Bong and others, 2020).

Objectivity as an abundance of copies: quantum Darwinism. According to the physicist Wojciech Zurek and his collaborators, a fact becomes objective in a precise case. It happens when in the surrounding environment there are very many copies of the same information. This abundance of copies is called **redundancy**. It is like when you make a hundred photocopies of the same sheet to hand them out to the whole school. Thanks to these copies, many observers can read that information. Each collects a different fragment of the environment. Nobody needs to know the information in advance. And nobody needs to touch or directly disturb the object (studies by Zurek 2003, 2009; Ollivier, Poulin and Zurek 2004).

The numbers at play in reality are gigantic. Take a tiny dust grain one thousandth of a millimetre wide (a micron). The grain is lit by the sun. It prints its own position in the light that bounces around it. It does so about a hundred million times in a single millionth of a second, that is, in a microsecond (Riedel and Zurek 2010).

How many copies must be collected to manage to read a fact? There is an approximate formula. It holds when the copies are many. Zwolak, Riedel and Zurek found it in 2014. The number of copies needed is about $\ln(1/\delta)$ divided by a certain quantity. This quantity is called the **quantum Chernoff information** of a single copy. Let us explain these pieces in words right away:

- δ (the Greek letter delta) is the **tolerance**. It is the small margin of error that we are willing to accept. For example, we can accept being wrong once in a thousand or once in a million.
- $\ln(1/\delta)$ is the **natural logarithm** of $1/\delta$. In simple words, it is a mathematical operation that counts how many zeros there are in the precision we ask for. The more precise we want to be (that is, the smaller δ is), the more this number grows. But it grows slowly. Doubling this number means asking for an enormously higher precision.
- The **quantum Chernoff information** measures how much a single copy helps us to tell apart the two possible values of the object. It is like asking: how sharp is a single photograph? If the photo is very sharp, a few are enough. If it is very faded, a great many must be added together to understand what is portrayed.

Let us now think of an ideal limiting case. Light reaches the object from all directions in exactly the same way, as inside an oven that is evenly lit. In this case the readable copies disappear completely (Riedel and Zurek 2011).

Physicists have also found the precise mathematical form of the quantum state that guarantees this objectivity. It is called the **spectrum broadcast structure**. In this situation every single fragment of the environment, taken on its own, carries the value of the system. And the fragments do not disturb each other (Korbicz, Horodecki and Horodecki 2014; Horodecki, Korbicz and Horodecki 2015; Le and Olaya-Castro 2019; review by Korbicz 2021).

There is, however, an important difference, discovered by Brandão, Piani and Horodecki (2015). **Observables** are *what* can be measured, that is, which question can be asked of the system. The objectivity of observables almost always arises by itself. **Outcomes**, instead, are *which answer* is found when making the measurement. The objectivity of outcomes arises only if there are suitable physical interactions between the object and the environment. These ideas have also been verified in the laboratory. Some experiments used photons (Ciampini and others 2018; Chen and others 2019). Others used a small defect inside diamond crystals, called an **NV centre**. This defect behaves like a tiny quantum compass (Unden and others 2019).

Existence as a matter of degree. In 2022 Zurek introduced the expression "relatively objective existence". The idea is this. The existence of a physical state is not a quality shut inside the object. It arises instead from the redundant copies scattered in the environment. And it is a matter of degree: there can be more or less of it. In the scientific literature, this is the starting point closest to what is proposed here.

Relative facts. There is another reading of the physics of quanta. It is called **relational quantum mechanics** and it was proposed by Carlo Rovelli in 1996. According to this reading, physical facts always exist *relative* to a system with which they interact. Facts become "stable" for another system thanks to a phenomenon called **decoherence** (Di Biagio and Rovelli 2021).

What is decoherence? In isolated quantum systems, the different possibilities can mix and produce **interference**. It happens as with waves on the water of a pond. When they meet, the waves add up or cancel out. When, however, the system touches the external environment (the air, the light), something happens. The interferences disappear from the behaviour of the system taken on its own. Its coherence is not destroyed. It is transferred instead into the links (the **correlations**) with the environment.

In Rovelli's theory one more rule is needed. It is needed to be sure that two different observers agree when they compare. This starting rule is a **postulate**, that is, a premise accepted without proof. It is called "**cross-perspective links**" (Adlam and Rovelli 2022). The rule says this. Suppose that observer A has recorded a result in a physical variable. If observer B measures that variable, B finds the same result as A. This theory has also received known objections from other scholars (Pienaar 2021; Brukner 2021). Brukner argues that "qubits are not observers". A **qubit** is the quantum bit, the simplest building block of quantum memory.

The theorems on Wigner's friend. Let us go back to the closed laboratory of Wigner's friend. Let us take certain assumptions that seem completely natural to us. Some recent mathematical theorems prove that they cannot all be true at the same time (Frauchiger and Renner 2018; Brukner 2018; Bong and others 2020). One of these assumptions is called by Bong and others the "**Absoluteness of Observed Events**". It says that the result recorded by an observer holds as a single and absolute fact for anyone. It holds also for whoever does not compare it.

These theorems have been put to the test in the laboratory with photons:

- Proietti and others (2019) measured a quantity called S. This quantity combines the links between different measurements. If all those natural assumptions were true together, S could never exceed the number 2. In the experiment, instead, they found $S = 2.416 \pm 0.075$. That ± 0.075 is the uncertainty of the instrument. Exceeding 2 by reaching 2.416 means going beyond the limit by more than five **standard deviations**. The standard deviation is the yardstick with which physicists measure the typical size of the random error. Exceeding it five times means that the excess is very clear. It cannot be a chance effect due to a tremor of the instrument.
- Bong and others (2020) showed that the limit predicted by "**Local Friendliness**" is exceeded. Local Friendliness is the set of those natural assumptions applied to Wigner's friend.

Other studies have classified all the variants of these theorems (Cavalcanti and Wiseman 2021). Others still have proposed to use an artificial intelligence as the "friend" in the laboratory. It runs on a quantum computer (Wiseman, Cavalcanti and Rieffel 2023). Moreover, Rivlin, Engineer and Baumann (2025) made a calculation. They looked at what happens if some partial copies of the friend's result escape from the closed laboratory. In that case the room for non-absolute events shrinks. In their formula the maximum limit becomes $2 + 4\varepsilon$. The formula is an inequality of CHSH type, similar to the one for S. Here $1 - \varepsilon$ is the probability of managing to guess the friend's result by looking at what came out of the laboratory. When the error ε is greater than or equal to 0.207, the limit is no longer violated. Watch out for a fundamental point. All these theorems on Wigner's friend concern events that are **not** compared between the observers.

The principle of agreement. In 1976 the mathematician and economist Robert Aumann proved a famous theorem. Let us take two subjects who start from the same basic information. Suppose that their estimates become **common knowledge**. This means that I know your estimate and you know mine. I know that you know mine, and so on to infinity, like two mirrors placed one in front of the other. Then the two cannot "agree to disagree". They must reach the same estimate.

This also holds when the two use quantum correlations. Disagreement would become possible only in hypothetical worlds. In those worlds the links would be even stronger than those allowed by quantum mechanics (Contreras-Tejada and others 2021). Leifer and Duarte (2026) proved that Aumann's theorem holds in any theory with a certain property. In the theory, the normal rule with which probabilities are added must work (the law of total probability). Cepollaro and Di Biagio (2026) added another point. To make the theorem work it is not even necessary to imagine a "single state of the world". It is enough that a **joint distribution** of the records exists. It is a single table of the probabilities of all the combinations of the records. The theorem can fail only where this single table cannot be written. This happens precisely in the Wigner's-friend scenarios that are not compared.

The common part in mathematics. Finally, in 1973 two mathematicians, Péter Gács and János Körner, gave the exact mathematical definition of the **common part** of two random variables. It is the richest possible quantity that can be computed in two ways. It can be computed starting only from the first variable. It can also be computed starting only from the second. In both ways one always gets the very same result.

2. The thesis of this work, explained step by step

Now let us leave the studies of other authors. Let us enter the heart of Ugo Lanciano's proposal, without using mathematical formulas.

2.1 What the thesis says

The thesis speaks about the **physical objects that we observe**. Think of a chair, a tree or a stone.

Usually we think of the object in a certain way. We imagine that it is a thing that is already out there, in a space ready to receive it. Then someone arrives and sees it. Here it is proposed to look at things the other way round. For an observer, an object is not a ready-made thing in a given space. It is a part of the **observation matrix** of the observer itself. This part is **transposed** and **projected** into smaller parts.

Let us stop right away to explain these three key words:

- **Observation matrix:** it is a term of the author's own. The AI gives its own reading of the author's theoretical framework. In this reading the matrix is the structure of counting, seen from the observer's point of view. The structure is made of **multiples**, that is, counting in a row (1, 2, 3...). It is also made of **fractions**, that is, dividing into parts (halves, thirds, quarters...). Think of a sheet of squared paper through which you measure everything around you. In the mathematical model of this work, the matrix is represented by the **record** of everything the observer has collected. It is a choice made by the AI.
- **Transposed:** it means that that part of the matrix presents itself to the observer as something that is elsewhere. It is at a certain distance from whoever looks. It happens as when the image on the cinema screen seems far away in front of you.
- **Projected:** it means that of all that richness only a simplified outline is kept. We will see it very well shortly, in section 2.2.

Why, then, do two different observers say that they see "the same object"? The author answers like this. Their matrices are not identical in every respect. But they are equal in large percentage.

In the model used in this work, what the two records have in common is their **reading** of the object. This reading is worth about 1 bit, accepting a very small tolerance. And here is the central point. The common object **is** precisely that common part. It is not a separate thing that lies hidden beneath the records and produces them. Be careful, though. In the model we can also look at the whole raw record, which is a choice of the AI. Then the common part between the two whole records is small (we will see it in chapter 4). So the model does not put a sentence of the author to the test. It is the sentence according to which the whole matrix is «uguale in larga percentuale» ("equal in large percentage").

To avoid confusion, let us fix right away the precise meaning of three words. We will always use them:

- **Observer:** it does not mean a human mind or a conscious person. It is any physical system able to record data and compare them. It can be a camera, an electronic sensor, a laboratory notebook. It is the same practical definition used in the experiments on Wigner's friend. There an observer is a system that extracts information and stores it in a physical memory (Proietti and others 2019).
- **Record:** it is everything the observer has noted down, understood as the physical state of its memory. For example, it is the pixels exposed on the sensor of the camera.
- **Reading (or projection):** it is what the observer obtains from its own record when it uses it to answer a simpler question. For example: "is the chair on the right or on the left?". It is an operation made on the record. Usually it throws away a large part of the useless details.

Two important things not to confuse. Someone might ask: "But then doesn't the chair exist on its own?" Let us make things clear. The chair, too, is a part of the overall structure. It has a state of its own. It also has its own local relations with what touches it, like the air of the room and the light that bounces off it. In this sense the chair is an autonomous entity. The thesis does not deny it at all.

The thesis says a precise thing instead:

1. What of the chair exists *for observer A* is the common part between the state of the chair and A's record. When the copies are imperfect, it is the common part computed with a very small tolerance. We will see it in chapter 4.
2. What *A and B agree on* is the common part between A's record and B's record.

Here are the three common parts at play:

- the one between the chair and A;
- the one between the chair and B;
- the one between A and B.

Suppose that A and B have collected enough copies, that is, enough grains of light. Then these three common parts

become exactly the same quantity. This is precisely what it means to say "A and B see the same chair".

Let us look at the numbers of the model (chapter 3). Let us take the case in which each observer collects 23 copies. Each single copy has a 10 % probability of being wrong. Then A's reading differs from the state of the chair with a tiny probability. This probability equals $4.7 \cdot 10^{-7}$, that is, fewer than 5 times in ten million. B's reading differs from A's with probability $9.4 \cdot 10^{-7}$ (about 9 times in ten million). In this scheme no part has a special or privileged place. The chair, A and B are three parts of the structure. The link between any two of them is measured with exactly the same rule.

2.2 The key word: what is a "projection" (the shadow, the ruler and the non-local phase)

We want to understand the word **projection** thoroughly. To do so we use three concrete images from everyday life.

First image: the shadow on the wall. Imagine putting your hand in front of a lit lamp. A shadow appears on the wall. The shadow keeps the outline of the hand, but loses everything else. You do not see the colour of the skin. You do not see the lines of the palm. You do not see the thickness of the fingers. Moreover, two different objects can cast the very same shadow on the wall. Think of a cardboard cut-out and of a real hand. A projection always does this job. It takes something very rich in details and brings it onto something poorer. Along the way it loses a lot of information.

Second image: the record is very rich, the reading is its shadow. Think of what happens in the physical record of a person (or of a video camera) looking at a chair. That record contains an enormous amount of details that belong to it alone. For example:

- which single photons arrived and at which precise instant;
- the tremor or the electronic disturbance of its instrument (the **noise**);
- the exact angle from which it is looking.

Now think of a second observer sitting on the other side of the room. Its record contains just as many details, but they are all different. Imagine taking the two complete records and comparing them line by line. The two records would hardly resemble each other at all.

To see the chair, each of the two observers makes a projection. It brings its own enormous record onto a much simpler question, for example "where is the chair?". And it throws away all the rest. It does like the shadow on the wall, which throws away the colour and the inside of the body. It is in this projection, and only here, that the two records coincide (in the model, within the chosen tolerance). The two shadows have the same outline.

The common object is what appears in the two projections with the same value. In mathematics, the **Gács–Körner common part** is exactly this. It is the richest projection that two records share. The copies can contain errors. In that case one uses the version with a very small tolerance (chapter 4). This is done because otherwise the exact common part would be zero.

The numbers of chapter 4 make this difference very clear:

- When each observer reads 23 copies, the common part is worth about **1 bit**. The question "where is the chair?" (on the right or on the left) is worth 1 bit. The record holds 99.9998 % of it. So almost the whole projection is in common.
- But how much does this single bit weigh compared with the whole raw record? At 23 copies, the whole record contains about 11.8 bits of data. They are all the details on the single copies and their errors. So the common part is only **8.5 %** of the whole record.
- The observers can go on collecting copies until they reach 61 copies. Then the whole record becomes even larger. The common percentage drops to **3.4 %**. The projection ("where is the chair?"), instead, stays almost entirely in common.

We could demand that two observers have identical whole records in order to say that they see the same thing. Then they would never see the same object. Let us always remember the correction seen at the beginning. These numbers (8.5 % and 3.4 %) concern the raw record of the model. In that record each copy carries its own noise. This is a choice of the AI and does not represent the author's "matrix". For two nearby observers, the author says that the matrix is «in larga percentuale uguale» ("equal in large percentage"). This model does not put that sentence to the test.

Third image: the ruler and the fractions. Measuring the position of a point with a ruler is another perfect example of projection. When you lay the ruler down, you do not read the infinitely precise position of the point. You only see in which mark it falls. That is, you read a **fraction** of the unit of measurement. And you lose the detail smaller than the

mark.

Now imagine two observers who have two different rulers to measure a bar one metre long:

- The first has a ruler divided into **sixths** (6 equal parts).
- The second has a ruler divided into **ninths** (9 equal parts).

Their readings of the same position will be different, because they use different marks. However, both rulers share some marks. They are the **thirds**: in fact both 6 and 9 are divisible by 3. The two observers can limit themselves to saying in which third the object falls. Then their readings always coincide. This is why the author speaks of projection "in fractions". Each reads the world using the fractions of its own scale. And what two observers have in common lives in the fractions common to their two rulers.

What casts the shadow? The non-local phase. When we see a shadow on the wall, we know that there is a solid body producing it. The body is placed in space. But what is there "underneath" the projections of the records in the author's theoretical framework?

The author puts a **non-local phase** at the basis. It is made of relations that exist without space and without time. Pay attention to the words. Here "non-local" does not mean a magic force that travels at a distance. Nor does it mean a signal faster than light. It means instead that the relation does not live in a precise place. It is present in the same way in every place and at every instant.

Let us now think of the **local** world. It is the world made of space and time in which we move every day. What is this world? It is that same reality, but counted from a point of view. They are the copies of a relation. Whoever reads the copies arranges them in a row, one after the other, and this is **time**. Or arranges them side by side, and this is **space**.

Now let us look at a reading proposed by the AI. The reading is not proven. In fact, identifying the copies of the theoretical framework with the photons of light is an interpretive choice. In this reading, every single photon that bounces off the chair is a copy of the same relation. And between two copies of the same thing there is no distance. Read this way, this is the reason for a fact. Two people see the same chair while collecting completely different photons. Their two records are two counts of one single non-local relation. They are made from two different points. And the common part of the records is the trace of that single relation in the two counts. This is how the theoretical framework gives a reason why the two shadows coincide. Underneath, the relation is only one.

In the author's framework, the observed object thus arises as a **non-local transposition and projection of the observation matrix**. The observer projects its own matrix and discards the private details. What does it have left in its hands? Not a datum shut inside its head or its memory. It has left the trace of the non-local relation itself. This relation has no distance between its copies. This is how the framework reads a precise fact. A's projection coincides both with the state of the chair and with B's projection. In the model this holds except for a tiny probability, of the order of one in a million, seen in section 2.1.

The author's framework uses exactly the same idea to give a reason for **quantum entanglement**. Entanglement is the phenomenon by which two very distant particles show linked results when they are measured. The two distant measurements exchange no message through space, because they are simply the two ends of a single relation. And this is also how the hardest and most surprising side of the thesis is read. What has no non-local link with us casts no shadow in our record. And so *for us it does not exist*.

What is lost in the projection. Think of the passage from the whole record to the reading. Everything that is discarded is not an error or a defect of the instrument. It is simply the private part of the record. It describes the particular point of view of whoever looks. Two very different records can give the very same projection. It happens as with two different objects, which can give the same shadow on the wall. This is why the common object is not the whole content of the records. It is only what remains identical after making the projection.

2.3 Three reasons to think of it this way

Why is it worth adopting this point of view on objects? There are three precise reasons:

1. **One always goes through the record.** No observer can ever touch or know an object except through its own physical record. It is not a philosophical idea. It is a practical fact. Think of anything we know about a star, an atom or a table. It is written in the physical state of our eyes, of our instruments or of our memory.
2. **Modern physics has already moved in this direction.** In Einstein's theory of **relativity**, let us take two distant events. Saying that they happen "at the same instant" (**simultaneity**) does not hold for everyone. It depends on

how the observer moves. What everyone shares are only two things. They are the coincidences at the same point and the **causal order**, that is, which event can be the cause of another. In quantum mechanics, the state is relative to the observer (Rovelli). The results inside a sealed laboratory are not absolute for whoever is outside (Brukner; Bong and others). And objectivity arises from the copies scattered in the environment (Zurek). Here one more step is taken along the same road. It is said that the *observed object itself*, too, is what the records have in common. Not only its state or its properties.

3. **The author's starting framework asks for it.** This work arises from a **premise** (a postulate) of the author. An entity exists for another exactly to the extent of what they have in common. And at the same instant the second also exists for the first. If existing means having a relation, then the observed object is also a relation. What two observers have in common is what exists for both of them. Moreover, the framework distinguishes two phases, as we saw in section 2.2. The non-local phase is made of relations without space and without time. The local phase is the same reality counted from a point. The author's original words are kept in a separate file (`AUTHOR_BACKGROUND.md`).

2.4 How this thesis reads four known facts

Let us see how this point of view gives a clear reason for four phenomena that we know:

- **Why compared records are never in disagreement.** In the ideal case, what does "comparing two records" mean? It means that observer B makes a faithful copy of observer A's reading. From that precise instant, that reading becomes part of their common part. And on the common part it is impossible to disagree. The very meaning of the word "common" says so. Adlam and Rovelli ask for the "cross-perspective links" as a postulate. Here these links become the ideal case of comparison. The measure of the common part, instead, tells us how much A and B already shared *before* talking to each other. Be careful: this is not a mathematical proof of Adlam and Rovelli's postulate. It is the way in which the thesis reads it.
- **Why the reading inside a sealed laboratory is not absolute.** In the Wigner's-friend scenario there are two possible situations. First case: no copy of the friend's result comes out of the closed laboratory. Then Wigner's record outside does not depend on that result. The common part with the friend's reading is exactly zero. In our model this is a basic check, true by construction. Second case: Wigner, from outside, uses his quantum instruments to undo the operation carried out in the laboratory. Or he makes an interference measurement on the whole laboratory taken together. In this second case quantum mechanics does not even provide a single table of the probabilities of the two results. This table is a joint distribution. So the common part cannot even be defined. It is the very same point at which Aumann's agreement theorem can fail, according to Cepollaro and Di Biagio. In both cases, the thesis does not declare absolute the reading of the friend shut inside. So it is perfectly consistent with the theorems on Wigner's friend. These theorems call into question precisely the absoluteness of the events that nobody compares.
- **Why in everyday life things seem to us to exist "all or nothing", even though existence has degrees.** Let us take the model with a 1-bit object and a fixed tolerance. Here the common part jumps from zero to the whole object within a few tens of copies. An ordinary object, like a dust grain in the sun, leaves hundreds of millions of copies every millionth of a second. So it is enough for us to collect a very small fraction of that light. And we immediately find ourselves well beyond the threshold. Let us think instead of an object with more possible values and of locally blurred copies. In this case the common part grows in steps, within the class of readings tested. The steps, however, do not follow the divisors of the grid. It is the calculation of chapter 7.
- **Why we see objects at a certain distance from us.** The distance of an object is read by comparing two different points of view of mine on the same object. For example, I look at it with the right eye and with the left eye. This is **triangulation**. In this way the primary distance is the one from the observer. The distance between two different objects is derived from those, instead. Triangulation, however, is ordinary geometry. It does not translate the "fractionation" the author speaks of. There is another way to compute the distance. One counts what fraction of the emitted copies we have managed to collect. The farther we are from a light bulb, the less light enters our pupil. This way translates the idea of fractionation more closely. But to work it needs to know how many copies in total were emitted by the source.

2.5 The price to pay (said plainly)

Every clear scientific idea has consequences that must be accepted fully. Here is what this thesis entails:

- **There is no "object in itself" detached from all records.** Think of something that has no non-local link with us. It is not an object hidden somewhere. Simply, *for us it does not exist*. In the model, the common part of the records is a clue to that non-local link. It is not its definition. This is the side of the thesis that clashes most with our mental habits. This side looks paradoxical. And it is precisely what makes the thesis different from traditional views.
- **Existence for another becomes a matter of degree.** It is no longer only "exists" or "does not exist" in an absolute sense. In the 1-bit model the degree is almost always zero or full (chapter 4). But in principle

intermediate situations exist.

- **It is not philosophical idealism (it does not depend on the mind).** In philosophical idealism one thinks that the world is created by human thought. Here there is nothing of the kind. Observers are not minds but physical systems. Records are physical memories. And the common part is a measurable physical correlation between two records. Moreover, the relation is **reciprocal**, that is, symmetric. If A has a common part with the object, the object has exactly the same common part with A.

2.6 What is new compared with the other studies

Now let us compare this thesis with the works of other scientists and philosophers. In this way we understand what different thing it adds.

A search was made in the abstracts of scientific papers (on arXiv) and in the texts cited. As far as was found, no other studies with this idea turn up. The idea is to define the *observed object itself* as the common part of the records of two observers. Of course this search cannot prove that none exist at all.

Let us look one by one at the differences from the closest studies:

- **Compared with Zurek (quantum Darwinism):** Zurek starts by taking for granted that there is a physical system chosen at the start. This system has its **pointer states**. They are the stable states that withstand the contact with the environment and are copied. Think of a coin lying on the table, which steadily shows heads or tails. It is different from a coin spinning on its edge. Around the system there is the environment that carries its copies. For Zurek, existence is the redundancy (the abundance) of these copies of the system. Here, instead, one does not choose at the start who is the "system" and who is the "observer". The common part is defined from the records of any two parts. The chair itself is also one of these parts. Moreover, the relation is perfectly symmetric, also between the object and the observer.

- **Compared with Rovelli, and Adlam and Rovelli (relational quantum mechanics):** they say that facts are relative to single systems. Moreover, they add the agreement between perspectives as a starting rule (postulate). Here, instead, agreement becomes a quantity that is computed with numbers.

- **Compared with Leifer and Duarte (2026):** at the end of their paper they make a proposal. They openly declare that it is a purely speculative hypothesis. They suggest that a shared objective reality emerges from common knowledge. It is the idea closest of all to the thesis. Here that idea becomes a precise definition: the common part of the records. And it is put to the test on a model.

- **Compared with Healey's objection (2013):** the philosopher of physics Richard Healey raises an important objection to quantum Darwinism. **Consensus** is not **verification**. Many people who agree do not make true what they agree on. For example, ten classmates may all copy the same wrong homework. They will all give the same answer. But the answer remains wrong. The thesis of this work keeps well apart the two things seen in section 2.1. On one side there is the common part between A and B, that is, what they agree on. On the other there is the common part between the chair and each of them. These two things coincide when the two read independent copies above the threshold. The two, instead, might read the same defective copies. Then they would agree on something that is not what of the chair exists for each of them. It is precisely the consensus without verification that Healey speaks of. The thesis recognises it as such.

- **Compared with ontic structural realism (Ladyman and Ross 2007):** this philosophical current holds that relations come before objects. The thesis offers a precise and measurable case of it. For two parts of the structure, the object is a relation between their records. And its size is computed in bits.

- **Compared with QBism (Fuchs, Mermin and Schack 2014):** QBism is an interpretation of quantum mechanics. In this interpretation the probabilities and the results of measurements are personal. They belong to the single subject (the agent) who makes the choice. Here, instead, there are neither agents nor personal probabilities.

- **Compared with Aumann and Contreras-Tejada:** they speak of agreement between rational agents. The mathematical structure they use is called the "**meet**" **of the partitions**. A **partition** is a division into groups. Think of how the cards of a deck are divided by suit or by colour. The "meet" is the finest subdivision that both subjects can recognise. Each recognises it using its own information. When the number of possible outcomes is finite, this structure corresponds to the Gács–Körner common part. It is a correspondence of mathematical structure, not of concepts.

3. How the thesis was put to the test

We are about to enter the virtual laboratory of this study. We will see three things:

- how the model used for the calculations is made;
- which are the eight mathematical translations of the idea;
- which rules they had to pass.

3.1 The model used for the calculations

The model is of the same type as those used in quantum Darwinism. Let us imagine a quantity X of an object. This quantity can take only two values. For example, two positions: position 0 or position 1. Each of the two positions has probability $1/2$, that is, 50 %, like heads or tails. This means that the object carries exactly **1 bit** of information. In fact, a bit is precisely the information contained in the choice between two equally likely possibilities.

Around the object there are N **carriers**, that is, N scattered photons. They are grains of light that bounced off the object. Each one carries a copy of it around. Light can hit an obstacle or bounce off an object. Then it spreads out into the environment. It is a phenomenon linked to **scattering** and **diffraction**. It is the way in which light waves open out and bend when they meet the edges of a body. It happens as with sea waves when they pass through the mouth of a harbour.

The model was studied in two versions:

1. **Classical version:** each copy travels towards the observers but can be wrong with a certain probability ε (epsilon). Being wrong means saying 0 when the object was in 1, or vice versa. It is like a somewhat absent-minded messenger who reports the opposite once in ten times.

1. **Quantum version:** the overall state of the object and of the N photons is written with the formula $(|0\rangle|e_0\rangle^{\otimes N} + |1\rangle|e_1\rangle^{\otimes N})/\sqrt{2}$. Let us explain it in words right away. The two possible values of the object, 0 and 1, each leave their own imprint (e_0 or e_1) on all the N photons. Then there is an important number called the **overlap** $s = \langle e_0|e_1\rangle$. It is a number between 0 and 1 for each carrier. The overlap s measures how similar the two imprints left on a single photon are. And so it measures how hard it is to tell them apart:

- if $s = 0$, the two imprints are completely different. It is like a red stamp and a blue one: a single photon is enough to understand at once where the object is;

- if $s = 1$, the two imprints are identical. The photon does not help us at all to tell 0 from 1.

Two observers, A and B, collect and read m_A and m_B carriers respectively. Their **reading** is the best possible estimate they can make with the photons collected:

- in the classical case, they count how many copies say 0 and how many say 1. Then they choose the **majority**, as in a class vote;

- in the quantum case, they use the **Helstrom measurement**. It is the best procedure allowed by the laws of quantum physics. It serves to tell two states apart with the least risk of being wrong.

In the model, the fact that space has three dimensions is taken as a starting datum.

What the model uses and what the definition uses. Here there is a very important distinction to understand. The model uses a source X and its photons at the start. It does so because in physics this is how the links between records are formed. They are formed through the light scattered by an object. But the **definition of the common object** does not look at the source. It uses *only* the joint distribution of the two final records, written $P(R_A, R_B)$. It is the table of the probabilities of all the possible pairs of records. First the records are formed. Then the common part is computed by looking only at that table, without knowing absolutely anything about the source. This respects criterion C4, which we will see shortly. So the thesis does not deny the physical interactions that fill the records. It denies, instead, that there is any need to imagine an additional "common object". The parts of the structure and their common parts are enough.

Three small additional models without photons. Two formulations and one check do not use the photon model:

- Formulation **T-3** uses two grids divided into a and b intervals. The grids are drawn on the same segment, exactly like two rulers with different marks. It serves to study the common part in its simplest form, that of numbers.
- Formulation **T-7** uses the geometry of spheres and triangles in three-dimensional space. It serves to study distance.
- The check with **moving observers** uses events in the spacetime of Einstein's special relativity.

3.2 The eight mathematical formulations put to the test

Each formulation tries to translate into mathematical formulas three expressions used by the author. The expressions are kept in the background file:

1. the "**matrix**" of an observer. Here, by a choice of the AI, it is rendered by its raw record;
1. the "**projection**" (which here is the reading, explained in section 2.2);
1. the part «**uguale in larga percentuale**» ("**equal in large percentage**") (which here is the common part).

Here is the table of the eight formulations (from T-0 to T-7) explained in a simple way:

Code	Name of the formulation	What it says in one simple sentence
T-0	No new consequence	It is the starting reference: ordinary quantum mechanics with the addition of the cross-perspective links.
T-1	The same matter	The common part is the single photons that were read *both* by A *and* by B (that is, the grains of light physically shared).
T-2	The copies of the same fact	Two copies that come from the same fact count as the same fact; it is enough that each observer collects enough of them.
T-3	The common divisor	Each observer measures with a ruler divided into a (or b) equal parts; the common part is the ruler with the common marks.
T-4	The common part of the records	It uses the mathematical common part of Gács and Körner, in three variants: **exact** (**T-4s**), or with a **tolerance** δ (**T-4δ**); plus a **graded measure** (**T-4g**), that is, the **mutual information** between the readings (which measures how linked the two readings are to each other, even though it is not itself a common variable).
T-5	The chains of copies	Two observers have a common object if there is a chain of copies that connects them (even only by a thread).
T-6	The same pattern	The common part is simply how similar the statistics (the shapes of the distributions) of the two readings are.
T-7	The distance	It computes the distance of the object: either by looking at what fraction of the total copies was collected (**luminosity**, **T-7L**), or by comparing the directions from two points of view (**triangulation**, **T-7P**).

3.3 The seven check criteria (fixed before doing the calculations)

An honest check was needed. So, before running the programs on the computer, seven **criteria** were fixed (from C1 to C7). The formulations had to respect them:

- **C1 (Agreement on comparison):** when two records are compared, they must be in agreement. Let us remember

that in version 2.62 this criterion had a second part added by the AI. This part was removed in this version 2.62.1.

- **C2 (Consistency on the sealed laboratory):** it must be consistent with the position already published by the author on the Wigner's-friend experiments (in deposit MM 2.48). The position is this. The readings made inside a laboratory from which no copy comes out are not absolute.

- **C3 (Reciprocity):** the relation must hold in the same way in both directions (from A to B as from B to A).

- **C4 (No object given in advance):** it must be possible to compute the common part by looking *only* at the records. There must be no need to know the source at the start.

- **C5 (Distance from the observer):** it must allow the distance from the observer to be read. The three dimensions of space are taken as a datum.

- **C6 (Declared parameters):** if a number chosen by hand is used (like the tolerance δ), it must be declared openly.

- **C7 (Limiting cases):** in the extreme cases it must give the right result. If the two records are identical, it is like having a single observer. There can also be no link at all. In the model this case is rendered by two completely independent sources, like two coins tossed in different rooms. Then no common object must result.

The formulations were put to the test on nine concrete situations, the test cases. Here are some of them:

- two observers who collect different photons;
- the sealed laboratory;
- the direct comparison of the records;
- two independent coins;
- the calculation of distance;
- reciprocity;
- moving observers.

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4. What comes out of the calculations (the results)

Let us now look at the outcomes obtained by putting the eight formulations to the test on the models. Let us start from three examples so simple that they can be understood and checked even by hand.

Three examples that can be done by hand

1. **The two rulers and the Greatest Common Divisor (T-3).** Take a ruler divided into 6 equal parts and another divided into 9 equal parts. Which marks coincide exactly on both rulers? They are the marks of a ruler divided into 3 equal parts. The computer checked this calculation for all the $40{,}000$ possible pairs of grids from 1 to 200 intervals. The common part of two grids divided into a and b intervals is always exactly the grid divided into $\mathrm{GCD}(a, b)$ intervals. Here GCD is the Greatest Common Divisor of a and b.

1. **Two people who look at the same chair while collecting different photons (T-1).** Two people look at a chair in the same room. The photons that enter the eyes of the first person are not the same ones that enter the eyes of the second. The sets of carriers are **disjoint**, that is, separate. So the material overlap of the photons collected is zero. And yet, in the quantum model (with overlap $s = 0.9$), each can collect 63 different photons. In this case the two observers agree with a probability equal to $1 - 8.6 \cdot 10^{-7}$. This means that they agree almost 100 %, being wrong less than once in a million. This shows that the agreement does not arise from having shared the same pieces of matter. The reading "the same matter" (T-1) would say instead that the two have nothing in common. The author's criterion C1 concerns only compared records. So this datum does not exclude T-1 by way of C1. However, T-1 remains **non-conforming** to criteria C4 and C5.

1. **Two independent coins and the fall of the "same pattern" (T-6).** Imagine that I toss a coin in Rome and you toss another coin in Milan. Both coins give heads 50 % of the time and tails 50 % of the time. The two statistical

tables (the "patterns") are identical. So their similarity is worth 1, the maximum. But we are not looking at the same coin at all! Having the same statistics does not mean having a common part. So **the reading "the same pattern" (T-6) falls**. It falls because it is not able to tell two independent coins from two records that look at the same object.

In the 1-bit model, the common part with tolerance is like a "yes or no" switch

Let us now see what happens to the common part with tolerance (T-4δ). The object is worth 1 bit and the copies are wrong 10 % of the time.

Tolerance δ (accepted error)	**How many copies each observer needs for the common part to become about the whole object (≥ 0.9999998 bit)**
10^{-3} (1 in a thousand)	**11 copies**
10^{-6} (1 in a million)	**23 copies**
10^{-9} (1 in a billion)	**37 copies**

The observers may have collected a number of copies below these thresholds. Then the common part is **zero**. This holds within the class of readings tested. They are readings that answer 0 or 1. They are based only on how many copies say 0 and how many say 1. As soon as they go above the threshold, the common part jumps at once to almost 1 whole bit.

Of course the number of copies needed depends on how good the copies are. We can use quantum copies that are very hard to tell apart, with overlap $s = 0.99$. They are almost identical imprints. Then, to have an error of 10^{-6}, **619 copies** per observer are needed. But let us remember that the objects of everyday life leave hundreds of millions of copies every microsecond!

What happens instead to the **exact common part** (T-4s)? It is the one that does not allow even a millionth of tolerance. It is enough for the copies to have the slightest imperfection. Then the exact common part is **zero** until the two compare. It stays zero even when it is extremely likely that they agree. The comparison means that B faithfully copies A's reading. When the comparison takes place, the exact common part becomes **1 bit** at once. Read this way, T-4s says a precise thing. In the model, with imperfect copies, without comparison nothing is common in the strictest sense. And it is precisely the comparison that makes it common. This outcome is compatible both with the data and with the position expressed by the author in deposit MM 2.48.

What "large common part" means

Let us go back to the difference between the projection and the whole record:

- We can measure the common part on the **projection**, that is, on the part of the record that speaks about the object. Then it is very large. Already at 23 copies the record contains 99.9998 % of the information about the object (0.9999975 bit out of 1). And the common part between the two records is worth about 1 bit (at least 0.9999998 bit above the threshold). In practice, everything the record says about the object is in common.

- We can instead measure it on the **whole raw record** of the model. At 23 copies this record contains about 11.8 bits, counting all the single photons and their noise. Then the common part is small. It is **8.5 %** at 23 copies and drops to **3.4 %** at 61 copies. All the rest of the record belongs only to whoever holds it. Let us remember again that this raw record is a choice of the AI, not the author's matrix.

There is also a simple geometric calculation. Imagine that each observer manages to see all the objects within a sphere of radius R around itself. It is like the circle of light of a lantern in the fog. Imagine then that two observers are at distance D from each other. What fraction of the objects seen do they have in common? Let us assume that the objects are distributed uniformly and with a sharp threshold.

- If they are at the same point ($D = 0$), they share the fraction 1 (100 % of the objects).
- If they are at distance $D = R/2$ (half a radius), they share 0.46 (46 %).
- If they are at distance $D = R$, they share 0.19 (19 %).
- If they are $D = 2R$ apart or more, the two spheres do not touch and they share 0.

What counts is the quality of the copies collected, not how many there are around

In the quantum model a check was made. The total number N of photons scattered in the environment was varied from a thousand (10^3) up to a million (10^6). Well, the curve of the mutual information always reaches its **plateau** at exactly **65 copies**. The plateau is the flat stretch at the top of the climb. There, collecting more copies no longer adds anything, because almost everything about the object is already known. The calculation is made with $s = 0.9$. It is required that less than a millionth of a bit be missing, that is, a deficit of 10^{-6} bit.

This recovers in the calculations the formula of Zwolak, Riedel and Zurek (2014), which we met in chapter 1. The number of copies needed is about $\ln(1/\delta)$ divided by the quantum Chernoff information of one copy. For pure quantum states, this information is worth exactly $-\ln|\langle e_0|e_1\rangle|^2 = -2\ln s$. It represents the speed at which the squared overlap shrinks each time we add a copy. Let us put $s = 0.9$ and $\delta = 10^{-6}$ into the formula. The division gives about **66 copies**, very close to the 65 found by the program. We can be content with a lower precision, that is, with a deficit of 1 %. Then these are enough:

- **21 copies** when $s = 0.9$;
- **213 copies** when $s = 0.99$;
- **2140 copies** when $s = 0.999$.

The summary table of the outcomes for the eight formulations

Here is the complete table that shows the outcome of each formulation put to the test:

Formulation **Outcome of the test**

T-0 (no new consequence) Starting **reference**.

| | **T-1** (the same matter) | **Non-conforming** (it does not respect criteria C4 and C5). In version 2.62 it was called "excluded" because of a part of criterion C1 that had been added by the AI.

| | **T-2** (the copies of the same fact) | **Cannot be judged in the model**: identifying the author's "nexus" with the quantity of the source was a choice of the AI (in version 2.62 it came out as "not reciprocal and with the object given in advance").

| | **T-3** (the common divisor) | **Compatible** as an arithmetic rule of the common part: it is computed using only the joint distribution of the two readings (test P1). The criterion on distances (C5) is met only if it is used together with triangulation, exactly as for T-4 (in version 2.62 it was called "non-conforming").

| | **T-4s** (exact common part) | **Compatible**: in the model, with imperfect copies, it is zero without comparison and becomes 1 bit when B faithfully copies A's reading (in version 2.62 it was called "excluded" because of the part of C1 added by the AI).

| | **T-4δ** (common part with tolerance) | **Compatible**, within the class of readings tested; the tolerance δ is a declared parameter (criterion C6). The classification is provisional because it was tested on a specific model (copies with a 10 % error, equal fragments).

| | **T-4g** (graded measure: mutual information) | **Compatible**, understood as a measure of how correlated the two readings are with each other.

| | **T-5** (the chains of copies) | **Cannot be judged in the model**, for the same reason as T-2 (in the AI's translation, which identified the nexus with the quantity of the source, it would have given a common object even with a single copy full of noise; but that outcome assesses the AI's translation, not the thesis).

| | **T-6** (the same pattern) | **Excluded** as a definition of the object: it falls because it does not distinguish two independent coins from two records of the same object.

| | **T-7L** (distance from luminosity) | It computes **only the distance**, provided the total number of copies emitted by the source is known (a datum distinct from the mass, which in the model is not derived from it). The reciprocity criterion C3 does not apply to a distance.

| | **T-7P** (distance by triangulation) | It gives the distance knowing the baseline and the angles: it is **standard geometry**, which does not translate the author's "fractionation".

|

Moving observers and declared predictions that did not succeed

Check with moving observers. Let us take two observers who move at high speed relative to each other. They are inertial observers in Einstein's relativity. What do they have in common? They have in common the **causal order** of events, that is, who can influence whom. They do not have in common, instead, the simultaneity of distant events. The program recovers this known result of relativity on a sample of events. Let us take a speed equal to 60 % of that of light, $v = 0.6c$. Then the time order stays the same for **83 %** of the pairs of events. Of these, **50 %** are "timelike" pairs. They are pairs close enough in space to be able to send each other a signal slower than light. The general result is proven with formulas (analytic).

Predictions registered before the calculations that did not come true. For scientific transparency, some predictions made before the calculations are declared. They are the ones that gave a different number:

- Let us take copies that are wrong 10 % of the time and a tolerance of 10^{-6}. It had been predicted that **21 copies** would be enough. Instead, **23** are needed.

- It had been predicted to find intermediate values of the common part below the threshold. Instead, within the class of readings tested with a 1-bit object, there are none. One passes from 0 to 1 all at once. The intermediate values appear instead when the object has more possible values. They are the steps that we will see in chapter 7.

- Finally, there was a numerical calculation error in the first program. It was due to the computer subtracting two almost identical numbers. The error was corrected in a separate file. All the other calculations do not change.

5. What is not claimed (the declared limits)

Before going on, let us make clear everything that this work does **not** claim to do. In this way we do not confuse what was calculated with what remains outside.

- **The "measurement problem" is not solved.** Objectivity is not derived from the premises alone. And the great puzzle of quantum mechanics, called the "measurement problem", is not solved. It is a matter of understanding one thing. In the world of quanta a particle can be in several states together. But when we measure it we always see only one result. Why?

- **The statistics are the same as those of quantum mechanics.** In this study some definitions based on probabilities were tested. They are computed starting from the table of probabilities (the distribution of the records). This table is provided by quantum mechanics. In the cases examined, these definitions give no statistical prediction different from that of quantum mechanics. The common divisor and triangulation are arithmetic and geometric rules, not statistical ones. The thesis is an interpretation consistent with the known data. But it is not yet an experiment that makes it possible to tell this proposal apart from standard quantum mechanics.

- **The link with spectrum broadcast structures is partial.** Do you remember the "spectrum broadcast structures" of chapter 1? They are the state in which every fragment of the environment carries the value of the object on its own. Here the link is only this. Let us take classical records that are perfectly distinguishable with respect to a chosen scale. This scale is a **fixed basis**. It is like saying that we have decided whether to look at

"up/down" or at "right/left". In this case we look at the label that indicates which possibility occurred: the **branch**. The branch is a common variable of every pair of records. Spectrum broadcast structures ask for stronger mathematical requirements than this.

- **The link with Aumann concerns only the mathematical form.** As already said, the similarity with Aumann's theorem concerns the mathematical structure used, not the physical concepts.

- **The non-local phase is not put to the test by the calculations.** The non-local phase (without space and without time) is the author's interpretive framework. The computer calculations of this work do not put it to the test. The calculations show one thing. The reading "the common object is the common part of the projections" is consistent with the data. Underneath this common part there could be a non-local relation, and not only the ordinary physics of scattered light. Saying so remains an interpretation.

- **The chair does not disappear from the model.** It is not claimed that the source object (the chair) is erased from the model. In the model the source and its carriers serve to build the table of the probabilities of the records. Moreover, there is the "consensus without verification", that is, Healey's objection seen in section 2.6. To tell it apart from a true seeing, one needs precisely to compute the common part between the chair and each observer. The thesis treats the chair as one of the parts of the structure, with a state of its own. What is removed is something else. It is the idea of an additional "common object", given even before the records are formed.

- **Some of the author's words have no full translation in the photon model.** The photon model translates the idea of **projection** well. Three other key ideas of the author, instead, do not find a full translation within that model. They are the **matrix** made of multiples and fractions, the **nexus** and **fractionation**. This is why the formulations based on "copies of the same fact" (T-2) and on "chains of copies" (T-5) cannot be judged in the model. And this is why the common divisor (T-3) and triangulation (T-7P) remain two small separate models. One is made of arithmetic and one is made of geometry. In the classical model of chapter 7, however, we will see a link. The common divisor links up with the copies. It becomes the maximum ceiling or a stage of the common part.

- **Cases left for future studies.** Some cases were not calculated here. The first concerns records with very few copies disturbed by noise, in models other than this one. The second concerns the laboratory of Wigner's friend, when only some partial copies escape from it. This case was studied in quantum mechanics by Rivlin, Engineer and Baumann in 2025. But it was not calculated with the rules of our thesis. The third concerns laboratories with larger and more complex "friends". In those cases the thesis would provide a criterion to say how much the common object exists. But it would not yet give a different statistics.

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6. The contact with mass (and why it does not appear here)

Let us now see briefly what **mass** has to do with counting copies. Mass is how much a body "weighs" or how much inertia it has. Let us also see why in this model mass stays outside.

In the author's general theoretical framework, mass is tied to the **multiple**. The multiple is counting things "in a row", one after the other. That link is studied in another, separate work (record MM 2.64). And that counting in a row is not the number of photons emitted by a light bulb or by a light source.

In the model used here, the definition of the common object does not use mass. There is a precise reason. All the copies counted are taken at the very same instant, arranged "side by side". It is a feature of a **static model**. A static model takes a single photograph, frozen in time.

In physical reality, instead, the copies do not all arrive at the same instant zero. They accumulate as time goes by, as we saw in chapter 1. That is, they are arranged "in a row". There is a problem that remains open. In reality, can the projection be defined without bringing in the multiple?

Finally, let us remember one thing about the distance computed using luminosity (formulation T-7L). To compute it one needs to know the total number of copies emitted by the source. This number is a datum completely distinct from the mass. In the model it is not derived from the mass. In reality it depends on how much the object is lit and on how hot it is.

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7. An object with more values (the novelty of version 2.62.1)

(Note: the content of section 9 of the original has been moved to chapter 7 of this version. Sections 7 and 8 of the original describe programs and procedure. Programs and procedure are gathered in the final chapter.)

We are about to see a new case. The observed object is no longer a simple 1-bit choice of "heads or tails". Now it can be in many different positions. This section answers a limit pointed out by an external review. In the 1-bit model of the previous chapters, the common part jumped from zero to the whole object all at once.

What was tested in this new model

A classical model was built, of the same type as the classical version seen in section 3.1. This time the object is a position on a **grid** made of n cells in a row. Grids of **6 cells** and of **12 cells** were tested. It is a bit like a strip of squares in a board game. Each of the two observers, A and B, collects m independent copies of the position of the object. But these copies contain **noise**, that is, they are slightly imprecise or blurred.

The questions that this computer experiment wanted to answer are two:

1. When the copies collected increase, does the common part with tolerance arise **in steps**? And do the coarse fractions (like the halves) appear first, and then gradually the finer ones (the thirds, the quarters...)?

1. Do these steps follow exactly the **divisors** of the number of cells of the grid? This is what the rulers of the Greatest Common Divisor (T-3) did.

How do the observers read the grid? They use readings that divide the strip of cells into blocks. This division is a **partition**. And they answer by indicating the most probable block. For example, let us take 12 cells. Dividing them into two equal blocks means asking oneself: "is the object in the first 6 cells on the left or in the last 6 on the right?". With the 12-cell grid, the computer tried all the possible divisions into blocks of cells next to each other (contiguous intervals).

This set of divisions is called the **class of readings tested**. Only these block readings are tried, and not all the mathematical functions one can imagine. So the result we find is a **lower bound** on the common part with tolerance. It means that the common part is *at least* this. More complicated readings could in theory find a little more of it.

Three different types of noise were tested (each with a fixed value):

- **Locally blurred copies (bell curve):** each copy is blurred around the right cell by about one cell. It follows a bell-shaped distribution. It is very likely to fall on the right cell or on the ones right next to it. It is very rare to end up far away.

- **One-step error:** the copy gives the right cell 80 % of the time. **20 %** of the time it is wrong, moving by a single cell to the right or to the left.

- **Error towards any cell:** when the copy is wrong, it can jump at random onto any one of the other cells of the grid. It can also end up very far away. It does not prefer the nearby cells.

Here too the translations are declared choices of the AI. The "fractions" the author speaks of are translated by the partitions (the blocks) of the grid. The «venuta in essere» ("coming into being") of fractionation is translated by the growth of the common part. The common part grows as the copies collected increase. There is also a safety check. In the new program an object of only 1 bit (2 cells) is put in. Exactly the same thresholds as in chapter 4 come out. They are **11, 23 and 37 copies** for the tolerances $\delta = 10^{-3}$, 10^{-6} and 10^{-9}.

Previous studies on similar problems

The definition of common part used is always that of Gács and Körner (1973). It holds for any variable with separate (discrete) values. In 1975 Hans Witsenhausen proved a result on the *exact* common part, that is, without tolerance. This part is always zero when the noise allows passing from one cell to another without barriers. In technical terms, it happens when the joint distribution has a **connected support**. It is exactly what happens in our cases with noise.

Sometimes one must distinguish among many different hypotheses by collecting many copies. In that case the speed at which the error drops depends on the pair of positions that is hardest to tell apart. It is the pair that has the smallest Chernoff information (Leang and Johnson 1997). Moreover, at point 4 we will see a curious effect. There a little noise

before the ruler helps to see better than the marks. This effect resembles, by analogy, a technique used in digital audio and images. But it is not the same thing. The technique is called **dithering** (Gray and Stockham 1993). In dithering a very light hiss is added to make the shades smoother. It is not claimed that nobody has ever studied before an object with more values with the common part with tolerance. A complete historical search of the literature was not made.

The five results on the object with more values

Here are the five results that came out of the calculations, explained step by step:

1. Within the class tested, the common part really grows in steps. Let us take the **12-cell** grid, with copies blurred by about one cell. The tolerance is $\delta = 10^{-6}$, that is, we accept one disagreement in a million. Here is what happens:

- From 1 up to **81 copies** per observer: the common part within the class tested is **zero**. The observers do not yet manage to agree even on which half of the grid contains the object.
- At **82 copies**: the first step pops up! The two observers agree on the **halves**. They divide the grid into 2 parts and the common part is worth **1 bit**.
- At **88 copies**: second step, they agree on the **thirds** (3 parts: 1.585 **bits**).
- At **91 copies**: third step, they agree on the **quarters** (4 parts: **2 bits**).
- As copies keep being added, the divisions into 5, 6, 7... parts gradually appear. In this way one arrives at **101 copies**. There the observers see the **whole object**: all the 12 cells distinguished one by one (3.585 **bits**).

There is exactly one step for each new boundary added between the cells. In all there are **eleven steps**, concentrated in the space between 82 and 101 copies. If we change the tolerance δ, the whole staircase shifts:

- with $\delta = 10^{-3}$ (more permissive), the ladder runs from **31 to 48 copies**;
- with $\delta = 10^{-9}$ (stricter), the ladder runs from **136 to 155 copies**.

The stricter the agreement we ask for, the narrower the staircase becomes in proportion to the total number of copies. Let us now take the other type of noise, the one that is wrong by one step 20 % of the time. With this noise the whole ladder is completed within **2-4 copies** (from **19 to 23 copies** with $\delta = 10^{-6}$).

Let us now think of the objects of everyday life, which send out hundreds of millions of copies per microsecond. For them, what follows is a reasoned estimate and not a direct calculation. Suppose that the copies stay independent and that the noise and the tolerance stay fixed. Then we go through those few steps very quickly. We do it so quickly that in practice existence still appears to us as a "yes or no", just as in chapter 4.

2. The steps do NOT follow the divisors of the grid. Here there is an important discovery. Each time we go up a step, we add a dividing wall between two nearby cells. This wall is a **boundary**. Putting in a boundary has a "cost". It is the risk that A's reading and B's reading fall one on the left and one on the right of that boundary, getting it wrong. This cost is almost identical at whatever point of the grid we decide to put the boundary.

The computer checked all the possible divisions. One sees that the best choice is always to divide the grid into blocks **as equal to each other as possible**:

- The number of blocks can divide 12 exactly. This happens with **2, 3, 4, 6, 12** blocks: halves, thirds, quarters, sixths, twelfths. Then the blocks all come out with the very same width.
- But the number of blocks can also **not** divide 12. This happens with **5, 7, 8, 9, 10 and 11** parts. In this case too the steps appear, using slightly unequal blocks! For example, let us divide 12 cells into 5 parts as equal as possible. The program makes two blocks of 3 cells and three blocks of 2 cells ($3 + 3 + 2 + 2 + 2 = 12$).

So the exact divisors of 12 give only the "clean" steps, that is, those with blocks that are all equal. But they are not the rule that governs the climb. Moreover, when one goes up from one step to the next, the new blocks are not always

contained inside the old ones. Think of the passage from the halves (blocks of 6 cells) to the thirds (blocks of 4 cells). The new boundaries cut the previous blocks in half.

3. "Coarse fractions first" depends on the local blurring. Why at 82 copies do the halves (coarse fractions) appear first, and only at the end the single cells (fine fractions)? The computer made the counter-test on the 6-cell grid. It used the third type of noise. It is the one that, when it is wrong, jumps at random onto any cell, even a distant one. Well, with that non-local noise, the halves and the thirds are **never** the best common part. The first step isolates a single cell or divides into unequal parts. The coarse fractions arrive first only when the noise is **local** (blurring). This happens because with blurring it is easy to confuse two nearby cells. It is instead very hard to confuse the left half of the grid with the right half.

4. The common divisor of the two rulers is the maximum ceiling or a stage, depending on where the noise enters. Let us put the two rulers of chapter 3 (formulation T-3) together with the noisy copies. Imagine that A and B look at the same grid using two different rulers, with a and b marks. The pairs of rulers with **6 and 9 marks**, and with **12 and 18 marks**, were tested. The noise shifts by one step 20 % of the time. Here everything changes depending on whether the noise enters *after* or *before* laying down the ruler:

- **Case A: the noise enters AFTER the ruler.** Each copy already gives a mark of the observer's ruler. But every now and then it is wrong by one mark. In this case, as the copies increase, the common part rises. It rises until it reaches exactly the common ruler made of $\mathrm{GCD}(a, b)$ marks. For the pair 6 and 9 they are 3 marks, which correspond to 1.585 **bits**. For the pair 12 and 18 they are 6 marks, that is, 2.585 **bits**. And **there it stops for good**. Within the class of readings tested it is an **exact ceiling** that cannot be exceeded. One can also compute mathematically a bound valid for *any* possible reading, not only for the block ones. The calculation is made with equally probable positions and up to 60 copies. Even with this bound, for $\delta = 10^{-6}$ no reading could exceed that ceiling by more than a thousandth of a bit (0.0011 bits). With $\delta = 10^{-3}$, instead, the mathematical bound is too loose. It is not enough to prove that it is a ceiling for all possible readings. Here the common divisor (T-3) works precisely as the **maximum ceiling** of the common part with tolerance.

- **Case B: the noise enters BEFORE the ruler.** The object is on a very fine grid. Each copy carries its position slightly blurred onto a nearby cell. Only *afterwards* does each observer read it with its own coarse ruler. The tolerance is $\delta = 10^{-6}$. With this tolerance the common part reaches the common ruler (GCD) at **21 copies**. It stays still there for a long stretch, up to over a hundred copies. But then **it goes beyond it**. At **115 copies** it reaches the precision of A's ruler. At **121 copies** it reaches that of B's ruler. At **133 copies** the observers manage to distinguish even all the cells of the very fine grid underneath! How is this possible? Think of the object when it is near the edge of a coarse mark. Then the blurring makes some of the copies spill over into the next mark. Each observer counts how many times the copies spill over. In this way it manages to understand where the object is even *inside* the mark of its own ruler. In this second case, the common divisor is not the final ceiling. It is a long **intermediate stage**.

5. What the other measures do. With the object with more values too, the **exact common part** (without tolerance) stays stuck at **zero** as soon as there is noise. It happens just as in chapter 4. This was verified with 1 and 2 copies in all cases. The **graded measure**, instead, rises smoothly and continuously, without making any step. The graded measure is the mutual information between the two readings. On the 12-cell grid it is already worth 2.680 **bits** with only 10 copies. It reaches 3.5847 **bits** (almost the maximum) at 80 copies. At that moment the common part with tolerance $\delta = 10^{-6}$ is still stuck at zero within the class tested. The steps arise precisely from the strict requirement to agree almost always.

What all this means for the thesis (and what the limits are)

Let us sum up this new section:

- The reading "the common part arises in steps, coarse fractions first" is **compatible** in the model. It holds within the class of readings tested and when the copies have a local blurring.

- The reading "the steps are the divisors", instead, does not follow: the steps are not exactly the divisors of the grid. **This reading of fractionation falls**. But the thesis of the common part does not fall.

- The common divisor of the two GCDs joins the common part of the copies. It becomes its **ceiling** if the noise enters after the ruler. It becomes instead one of its **stages** if the noise enters before. Which of the two cases

translates the author's thought better? Establishing it is an interpretive choice that this work leaves open.

Limits of this test. The model used in chapter 7 is only classical (not quantum). The class of readings tested is restricted to block divisions. So the numbers found are a lower bound. The exception is the case of noise after the ruler, where there is also the mathematical upper bound. Finally, for each type of noise only one fixed value was tested. The precise number of copies of the steps depends on those values.

How the work was done

In this final section we gather all the practical information on how the study was carried out and checked. We will see:

- who wrote the programs;
- how the calculations were verified;
- where the files are in the package;
- how anyone can redo the very same calculations on their own computer.

Who wrote and checked the calculations (provenance). The author, Ugo Lanciano, is not a professional physicist. The mathematical formulations, the computer programs and the calculations were made by working together with a first artificial intelligence (Claude, by Anthropic). Then they were checked independently by a second artificial intelligence. It is `gpt-6-luna` via Codex, by OpenAI. In the files of the package it is also referred to as Astra. The check took place through two rounds of verification. During these two verifications, the second AI raised 25 observations (points). Of these, 24 out of 25 were accepted and corrected. One instead was rejected by showing an exact calculation. The verifier then withdrew that observation. The author's original sentences from which the whole work arose are gathered separately in the file `AUTHOR_BACKGROUND.md`.

The check that led to version 2.62.1. For version 2.62.1, the author asked for an independent check from another, separate instance of Claude. This instance did not know the conclusions of the first. The check found the undeclared translation choices of the AI. They are the choices corrected at the beginning of this text. Every single point was rechecked against the original files. All the corrections were then also verified by the second artificial intelligence (Astra). The detailed plan of the corrections, the verification reports and the answers are in the folder `04_controllo_262_264`. In particular, they are in the file `04_controllo_262_264/PIANO_CORREZIONI_v1.md`. Afterwards there was an external review of the corrected version. The review added two clarifications in the chapter on the limits ("What is not claimed"). It also suggested the study on the object with more values. This new part, too, was verified by the second artificial intelligence in two rounds. The part is gathered in the folder `05_oggetto_piu_valori`.

Registrations before the calculations and digital fingerprints (SHA-256). Nobody was to be able to change the rules of the game after seeing the results. To guarantee this, all the test protocols, the predictions and the programs were registered *before* running the calculations. Each protocol and each program was given the exact time (in the universal **UTC** format). It was also given a **SHA-256 fingerprint**. Time and fingerprint were registered before the run. What is a SHA-256 fingerprint? It is a code of 64 letters and numbers generated by the computer from the content of a file. It is like a fingerprint. Someone might change even just a comma or a space inside the file. Then the fingerprint would change completely.

The programs and how to reproduce all the calculations. All the programs are written in the **Python 3** language. They use only the standard library included in Python and the mathematical library `numpy`. They save their results in data files in **JSON** format:

- For the calculations of the first sections (1-bit model, rulers, distance, relativity) there are four programs. They take less than a minute in all on an ordinary home computer. They are `livello1_conti.py` (the main registered run), `correzione1_conti.py`, `correzione2_conti.py` and `correzione3_conti.py`.

- For the calculations on the object with more values (our chapter 7, section 9 of the original) there are three more programs. They are in the folder `05_oggetto_piu_valori` and require about eighty minutes in all. They are `oggetto_piu_valori_conti.py`, `correzione1_conti.py` and `correzione2_conti.py`.

- All the calculations can be redone automatically from beginning to end. One can also check that they coincide perfectly with the registered ones. It is enough to run the program `run_all.py` by typing the command `python run_all.py`. The program takes about eighty minutes. When it has finished checking that everything matches, it writes "ALL MATCH" on the screen, that is, "everything matches".

Package contents and related works. The complete description of the package is also in the English file `README.md`. It is in the sections "Package contents", "How to run" and "Related records". The package contains:

- the registered protocol;
- the three corrections;
- the outcome notes;
- the four programs with the results;
- Astra's two verifications;
- the registers with the SHA-256 fingerprints;
- the folder `05_oggetto_piu_valori`, with protocol, corrections, outcome notes, three programs with the results and two verifications by Astra;
- the folder `04_controllo_262_264`, with the plan of the corrections of 2.62.1 and the related verifications by Astra;
- the program `run_all.py`;
- the file `AUTHOR_BACKGROUND.md` with the author's words;
- the file `LICENSE.md` with the licences of use: the free licence CC BY 4.0 for the texts and the MIT licence for the code of the programs.

This deposit is linked to three other public documents of the author on Zenodo:

- **MM 2.48** (DOI `10.5281/zenodo.23352814`): it explains how the author's thesis is placed with respect to the Local Friendliness experiments on Wigner's friend. This document was used for criterion C2.

- **Status report** (general DOI `10.5281/zenodo.21802475`), in the addendum of 26 August 2026 (`QUANTUM_PLACEMENT.md`, §0). It contains the earlier reading of agreement as a number of copies above a threshold. This reading was then developed in this work.

- **Entry page of the theoretical framework** (DOI `10.5281/zenodo.22667516`): it presents the author's postulate 2.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary of difficult words and of MM words

Here is a final dictionary. It serves to find at once the meaning of the words used in the text. There are the technical words of physics and the words of Mathematical Materiality (MM) itself.

- **Absoluteness of Observed Events:** the assumption according to which what an observer has seen and recorded is a single and absolute fact. This fact would hold in the same way for anyone, even for whoever does not compare it.

- **Bit:** the unit of measurement of information. It corresponds to a choice between two equally likely possibilities, like heads or tails. An object that can be in two places with the same probability carries 1 bit of information.

- **Causal order:** the relation that establishes which event can be the cause of another and influence it. Let us take the pairs of events that can influence each other. For these pairs the order stays identical for all moving observers. It is not so for the simultaneity of distant events.

- **Common knowledge:** the situation in which two subjects both know something. Each knows that the other knows it. Each also knows that the other knows that they know it, and so on to infinity.

- **Common part (Gács–Körner):** the richest quantity that each of the two observers can compute by looking only at its own record. Each gets the same value as the other. Then there is the version **with tolerance** δ. In this version the two always get the same value, except at most with a small probability of disagreement δ.

- **Copy (or carrier):** a fragment of the environment that carries with it an imprint of the value of the object. In our model it is a single photon of scattered light.

- **Cross-perspective links:** the starting rule (postulate) proposed by Adlam and Rovelli in relational quantum mechanics. Observer A has recorded an outcome in a variable. If observer B measures that variable, B finds the same outcome as A.

- **Decoherence:** the disappearance of the effects of quantum interference in the behaviour of a system taken on its own. It happens when the system comes into contact with the environment. Coherence does not vanish into nothing. It is transferred instead into the links (correlations) with the environment.

- **Diffraction (and scattering of light):** the phenomenon by which light bends and spreads out into the surrounding environment. It happens when it hits an object or passes close to its edges. The light carries with it the imprint of the position of the object.

- **Fractionation:** the author's word for dividing the unit into parts (fractions). The world is read through these parts. In the models of the work the AI translates it in two ways. The first is the divisions of the rulers (T-3). The second is the partitions of the grid into blocks (chapter 7). Triangulation, instead, is ordinary geometry and does not translate it.

- **Greatest Common Divisor (GCD):** the largest whole number that divides two given numbers exactly without leaving a remainder. For example, the GCD of 6 and 9 is 3. In rulers divided into a and b parts, $\mathrm{GCD}(a, b)$ indicates the number of divisions of the common ruler.

- **Helstrom measurement:** the best measuring procedure allowed by the laws of quantum physics. It serves to tell two states apart with the lowest probability of error.

- **Joint distribution:** the single table that lists all the possible combinations of two (or more) records. It also indicates the probability of each combination.

- **Local Friendliness:** a set of natural assumptions applied to Wigner's-friend scenarios. Among them are the absoluteness of observed events and the absence of influences at a distance faster than light. The name also indicates the mathematical limits that follow from them for the links measured in the laboratory.

- **"Meet" of partitions:** each of the two observers can make some subdivisions of the world with its own information. The "meet" is the finest and most detailed subdivision that *both* are able to recognise.

- **Mutual information:** a graded measure, expressed in bits. It says how much information one record provides about the other record (or about the object).

- **Nexus:** the author's own word for the relational link at the basis of what is counted and observed. In the photon model it has no full counterpart. Identifying it with the value of the source was a choice of the AI.

- **Non-local phase:** in the author's theoretical framework, it is the level of relations without space and without time. This level is present in the same way in every place and at every instant. "Non-local" means "not located in space", and not "force or signal that travels at a distance". The local world (with space and time) is that same reality counted from a point of view.

- **Observation matrix:** the author's own word. In the author's theoretical framework the multiples and the fractions of the common unit form a matrix. The AI reads this matrix as the structure of counting seen from an observer's point of view. In the photon model, by a choice of the AI, it was represented by the raw record of the photons collected.

- **Observer:** any physical system (an instrument, a sensor, a memory) that records information and compares it. It does not require a conscious mind.

- **Ontic structural realism:** the philosophical current according to which in physical reality relations come before objects.

- **Overlap s:** a number between 0 and 1. It measures how similar the quantum imprints left on a carrier by the two possible values of the object are. With $s = 0$ the imprints are completely different and very easy to tell apart. With $s = 1$ the imprints are identical and indistinguishable.

- **Partition:** a division of a set or of a grid into separate blocks. In the chapter on the object with more values (chapter 7), it indicates a reading. The reading says in which block of cells the object is.

- **Plateau:** a French word that means "high plain". In graphs it indicates the flat stretch of a curve. Here it

indicates the point beyond which reading more copies no longer adds new information about the object.

- **Pointer states:** the particular states of a physical system that withstand the contact with the environment without mixing. They are also the states that are copied by the environment.

- **Premise (or postulate):** a starting statement from which a reasoning or a theory begins. In the author's framework there is a fundamental premise (postulate 2). It says that an entity exists for another to the extent of what they have in common, in a reciprocal way.

- **Projection:** the operation that passes from something very rich in details to something simpler. The simpler thing keeps only an outline of it, just like the shadow of a body cast on the wall. In this work it is the reading that an observer obtains from its own record by discarding the private details.

- **QBism:** an interpretation of quantum mechanics. According to this interpretation the probabilities and the outcomes of measurements are personal assessments and experiences. They belong to the single subject (agent) who uses them.

- **Quantum Chernoff information:** a number that measures how much a single copy helps to tell apart the two possible values of the object. For pure quantum copies with overlap s, it is worth exactly $-2\ln s$.

- **Quantum Darwinism:** the physical theory proposed by Wojciech Zurek. According to this theory a fact becomes objective when the surrounding environment carries a great many independent copies of it. Many observers can read these copies without directly disturbing the object.

- **Reading:** a key word used to indicate what an observer obtains from its own record. It obtains it by answering a simpler question. It is a projection of the record.

- **Record:** everything an observer has noted down, understood concretely as the physical state of its memory.

- **Redundancy:** the abundance of independent copies of the same fact scattered in the environment.

- **Spectrum broadcast structure:** it is the precise quantum state in which every fragment of the environment, taken on its own, carries the value of the system.

- **Tolerance δ:** the small probability of error or of disagreement that is accepted. For example, one can fix a tolerance $\delta = 10^{-6}$. It means accepting that the two readings may fail to coincide at most once in a million.

- **Transposition:** in the author's framework, it is the fact that a part of one's own observation matrix presents itself as an object located elsewhere. The object appears at a certain distance from whoever observes. In the geometric model the distance is read by triangulation.

- **Triangulation:** the ordinary geometric method for computing the distance of an object. One compares the directions (the angles) in which it is seen from two different observation points.

- **Wigner's friend:** a thought and experimental scenario of quantum physics. One observer (the friend) makes a measurement shut inside a laboratory completely isolated from the outside. A second observer is outside: it is Wigner, the "superobserver". Wigner describes the whole laboratory as a single quantum system. He can also undo from outside what happened inside.

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